

B. Sc. (Hons) Semester III Examination 2023-24
BOTANY
Paper No. BOB-301 : Plant Ecology and Physiology

Time: 3 hours

Full Marks: 70

Note: Answer five questions, selecting two each from Sections A and B and Question 1 which is compulsory. Figures in the right-hand margin indicate marks.

1. **(Compulsory)** Answer/define the following in not more than 50 words each: [$10 \times 1 = 10$]

- (a) Invasive species [cite: 16068]
- (b) Habitat Mutualism [cite: 16069]
- (c) Density [cite: 16071]
- (d) Ecological pyramid [cite: 16072]
- (e) Allogenic succession [cite: 16077]
- (f) Apical dominance [cite: 16078]
- (g) Hydathodes [cite: 16079]
- (h) Embolism [cite: 16080]
- (i) H^+ -ATPase [cite: 16082]
- (j) Emerson enhancement effect [cite: 16083]

Section-A

2. Define population. Give a detailed account of various characteristics of a population. [15]
[cite: 16085]

3. Write descriptive notes on any **two** of the following: [$2 \times 7\frac{1}{2} = 15$] [cite: 16091]

- (a) General soil profile [cite: 16087]
- (b) Universal model of energy flow [cite: 16088]
- (c) Raunkiaer's life forms [cite: 16089]

4. Write critical notes on any **three** of the following: [$3 \times 5 = 15$] [cite: 16093]

- (a) Adaptations in Xerophytes [cite: 16094]
- (b) Analytical characters of community [cite: 16095]
- (c) Grazing and detritus food chain [cite: 16099]
- (d) Phosphorus cycle [cite: 16100]

5. What is ecological succession? Describe various sequential stages of a typical hydrosere. [15] [cite: 16101]

Section-B

6. Give an account of the C_4 and CAM pathways of CO_2 fixation in plants. [15] [cite: 16103, 16105]

7. Write critical notes on any **two** of the following: [$2 \times 7\frac{1}{2} = 15$] [cite: 16110]

- (a) Nitrate assimilation in plants [cite: 16107]

- (b) Photophosphorylation [cite: 16108]
(c) Photorespiration [cite: 16109]
8. Write short notes on any **three** of the following: [3 × 5 = 15] [cite: 16115]
- (a) Membrane transport [cite: 16112]
(b) Phloem loading [cite: 16113]
(c) Structure of phytochrome [cite: 16114]
(d) Electron transport system in mitochondria [cite: 16116, 16118, 16121]
9. What are plant growth hormones? Give an account of physiological roles of gibberellins.
[15] [cite: 16124]
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External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>